

Quiz (Transactions and Locks II)

Instructions: For each problem, you must perform all of the following tasks correctly in order to get credit.

1. Circle all SQL commands that block.
2. Write an X through all commands that error.
3. If a command causes a deadlock, write DEADLOCK and stop the problem.
4. Write the output of each SELECT statement.

You will be allowed to use a computer and access the internet, but you will not be allowed to connect to postgres and run postgres commands.

Problem 1:

Session 1

```
1 CREATE TABLE t ( a INT );
2
3 VACUUM t;
4
5
6
7 VACUUM FULL u;
```

Session 2

```
1
2 CREATE TABLE u ( a INT );
3
4
5 VACUUM FULL u;
6 VACUUM t;
7
8 SELECT count(*) FROM t;
9 SELECT count(*) FROM u;
```

Problem 2:

Session 1

```
1 CREATE TABLE t (
2     a INT PRIMARY KEY
3     DEFERRABLE INITIALLY DEFERRED
4 );
5 BEGIN ISOLATION LEVEL
6 REPEATABLE READ;
7 INSERT INTO t VALUES (5);
8
9
10 INSERT INTO t VALUES (6);
11 SELECT count(*) FROM t;
12
13
14 COMMIT;
```

Session 2

```
1
2
3
4
5
6
7
8 BEGIN;
9 INSERT INTO t VALUES (6);
10
11
12 INSERT INTO t VALUES (5);
13 ABORT;
```

Problem 3:

Session 1

```
1 CREATE TABLE t(a INT UNIQUE);
2 CREATE TABLE u(b INT REFERENCES t(a));
3 BEGIN;
4
5
6 INSERT INTO t VALUES (9);
7
8 COMMIT;
```

Session 2

```
1
2
3
4 BEGIN ISOLATION LEVEL
5 REPEATABLE READ;
6
7 SELECT count(*) FROM u;
8
9 INSERT INTO u VALUES (9);
10 COMMIT;
```

Problem 4:

Session 1

```
1 CREATE TABLE t (a INT PRIMARY KEY);
2
3 BEGIN;
4 INSERT INTO t VALUES (1);
5 INSERT INTO t VALUES (2);
6 INSERT INTO u VALUES (2);
7
8
9
10 LOCK TABLE u IN ROW EXCLUSIVE MODE;
11 SELECT count(*) FROM t;
12
13
14 INSERT INTO t VALUES (3);
15 COMMIT;
16
17
18
19 SELECT count(*) FROM t;
```

Session 2

```
1
2 CREATE TABLE u (b INT NOT NULL);
3
4
5
6
7 BEGIN ISOLATION LEVEL
8 REPEATABLE READ;
9 INSERT INTO t VALUES (2);
10
11
12 INSERT INTO u VALUES (1);
13 SELECT count(*) FROM u;
14
15
16 UPDATE u SET b=5;
17 COMMIT;
18 SELECT count(*) FROM t;
```